



MSc in Computing in Applied Data Science and Analytics

Data Science Applications Project report

Name of project

Submitted by: student name, student number

Supervisor: name of supervisor

submission date

Declaration

I hereby certify that this material, which I now submit for assessment on the programme of study leading to the award of MSc in Computing in the Institute of Technology Blanchardstown, is entirely my own work except where otherwise stated, and has not been submitted for assessment for an academic purpose at this or any other academic institution other than in partial fulfilment of the requirements of that stated above.

Author: _____

Dated: _____

Data exploration techniques can be seen in the following pages:	<i>pages x to y and pages w to z</i>
Data preparation techniques are described in the following pages:	

Abstract

Abstract here

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Chapter 1

Business Understanding

1.1 Business Objectives

In business terms, state what to you want to accomplish when mining this data.

1.2 Assess the situation

If there are any issues with access to data or time constraints, mention then here. Also details risks if any. Otherwise leave out this section.

1.3 Data Mining Objectives

State your technical objectives for mining the data.

1.4 Project plan

See suggested project plan on Moodle and adjust to suit your own project.

An example of a table is shown in 1.1

Different citation styles: Cognitive ability tests were originally developed to identify low academic achievers [Jensen, 1981; Munzert, 1980]. The first such test measured general cognitive intelligence, g , as identified by Spearman [1904, 1927].

1. BUSINESS UNDERSTANDING

Table 1.1: Defining broad factors of intelligence

Factor	Symbol	Description
Fluid Intelligence	Gf	Ability to solve problems independently of knowledge learned.
Crystallised intelligence	Gc	Acquiring and organising knowledge and skills, and ability to use such knowledge in solving problems.
Visual processing	Gv	Ability to process and analyse visual information
Auditory Processing	Ga	Ability to process and analyse auditory information
Processing Speed	Gs	Ability to perform automatic cognitive tasks quickly (measure in minutes)
Reaction Time / Decision speed	Gt	Speed at which an individual can react to a stimulus, or make decisions (measure in seconds).
Short-Term Memory	Gsm	Ability to hold information with immediate awareness and use it again within a few seconds
Long-Term Retrieval	Glr	Ability to store and retrieve information over a longer period of time.
Quantitative Knowledge	Gq	Ability to understand quantitative concepts and relationships, and work with numeric symbols. This is a measure of mathematical knowledge acquired, as distinct from mathematical reasoning (Gf).
Reading-Writing	Grw	Basic reading and writing skills (considered by Cattell-Horn to be part of Gc)

Chapter 2

Data Understanding

2.1 Introduction

The length of this section will be dataset and project specific. Guidelines can be agreed with your supervisor.

2.2 Describe the data

For each attribute, give its description and data type. For numeric attributes, give mean, min, max and st dev; for nominal attributes with a few values, list the values. (For large datasets, you may be able to do the above for groups of attributes if appropriate).

2.3 Explore the data

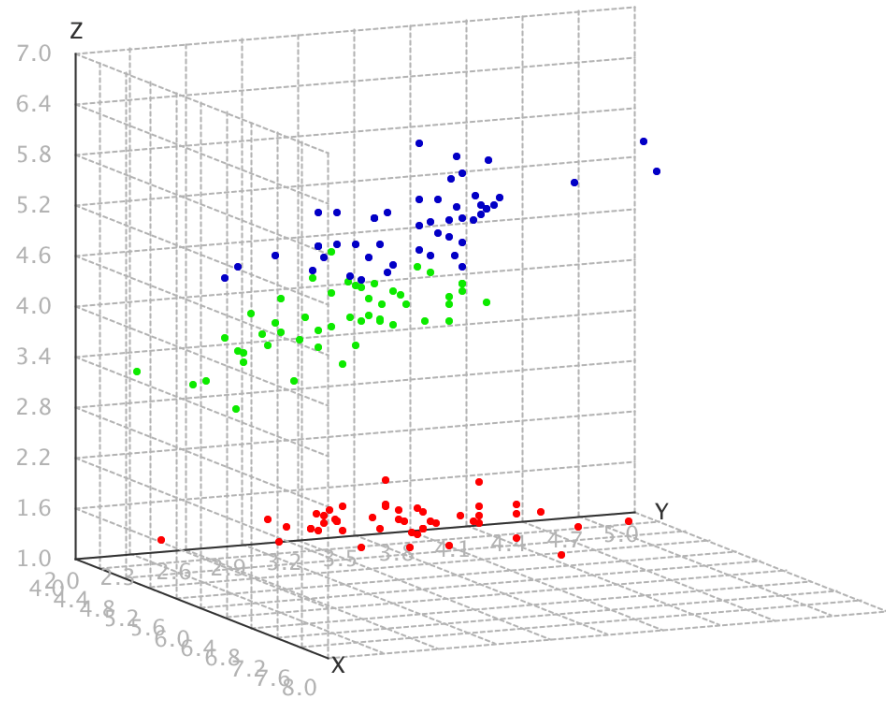
Discuss the results of an initial exploration of the data using graphs and exploratory statistics. You should NOT report on ALL attributes in this section, but comment on anything you found of note, such as attributes or groups of attributes that seem predictive; correlated attributes; attributes with limit value because of too much or too little variability, attributes with unusual distributions etc. Your discussion should be with respect to your initial business and data mining objectives. Your report should include an appropriate selection of data visualisations as covered in both Business Intelligence and Data Exploration.

2.4 Verify data quality

Does the dataset have many missing values? What type of missing data is it? The discussion should be detailed enough to inform how you intend handling the missing data. Is the presence of

2. DATA UNDERSTANDING

noise, bias or outliers likely to be an issue? Are there sufficient attributes and examples to achieve your mining objectives?



Text explaining diagram if needed

Figure 2.1: Scatter plot for Irish dataset

Chapter 3

Data Preparation

3.1 Introduction

The length of this section will be dataset and project specific. Guidelines can be agreed with your supervisor.

3.2 Select the data

If you need to reduce the number of rows or columns in the dataset, discuss the approaches you tried, and what worked best.

3.3 Clean Data

If data quality was an issue, discuss the approaches you tried, and what worked best.

3.4 Construct Data

Detail data transformations you tried, why you thought it would be useful, and how well they work. The report should get across the iterative nature of this phase, and show that you experimented with a selection of techniques as covered in data pre-processing lectures. It should also be apparent that you used the results of data exploration to inform this phase, rather than randomly trying different techniques in the hope that something would work. This section can be merged with the next section ? modelling ? if it makes it easier to link preparation techniques with the resulting model accuracy.

Note: CRISP-DM has an additional section here of integration, which will probably not be relevant for most of your projects, but if it is, you could include that here or as part of data selection.

3. DATA PREPARATION

3.5 Examples of equations

$$LOF_{MinPts(p)} = \frac{\sum_{o \in N_{MinPts(p)}} \frac{lrd_{MinPts} o}{lrd_{MinPts} p}}{|N_{MinPts(p)}|} \quad (3.1)$$

$$lrd_{MinPts(p)} = 1 / \left(\frac{\sum_{o \in N_{MinPts(p)}} reach-dist_{MinPts(p,o)}}{|N_{MinPts(p)}|} \right) \quad (3.2)$$

$$reach-dist_{k(p,o)} = \max\{k-distance(o), distance(p, o)\} \quad (3.3)$$

Chapter 4

Modelling

4.1 Introduction

Select modelling technique (1 paragraph approx) Discuss which algorithms are most appropriate for the dataset and mining objectives, justify your selection.

4.2 Generate Test Design

For classification tasks, discuss how you will generate training and test data., using cross validation, split validation or some other methods (1 paragraph) For all tasks describe how you will evaluate your results. (1 paragraph approx)

4.3 Build and Assess the model

For each algorithm: Detail the parameter values tried, the model generated (did you learn anything from the model itself, e.g. decision tree nodes). Discuss and interpret the model accuracy, and if relevant, how the accuracy might be improved. Include diagrams where relevant. (approx. 1 page per algorithm)

Key finding: example of a framed box

4. MODELLING

Chapter 5

Evaluation

5.1 Introduction

The purpose of this section is to document, in business, non-technical terms, what information has been learnt from the dataset. This discussion should focus on your original business objective(s), but can also include other things you have learnt along the way. Also in this section, discuss limitations of the study if relevant. For example, might model accuracy be unreliable because of small sample size, or bias in the sample; might quality issues have impact on model accuracy such as choices made to fill missing values or working with an unbalanced dataset.

5. EVALUATION

References

- Jensen, A. R. (1981). *Straight Talk about Mental Tests*. The Free Press (MacMillian).
- Munzert, A. W. (1980). *Test Your IQ*. MacMillian, USA.
- Spearman, C. (1904). General intelligence, objectively determined and measured.
- Spearman, C. (1927). *The abilities of man*. MacMillian.

REFERENCES

Appendix A

Descriptive name for Appendix

A. DESCRIPTIVE NAME FOR APPENDIX

A.1 Section 1

- Example of a bulleted list
- . . and item 2

Paragraph after forcing a space

Appendix B

Additional Data Exploration

1. Figure 1 is referenced in chapter
2. Figure 2 illustrates xx as discussed on chapter y

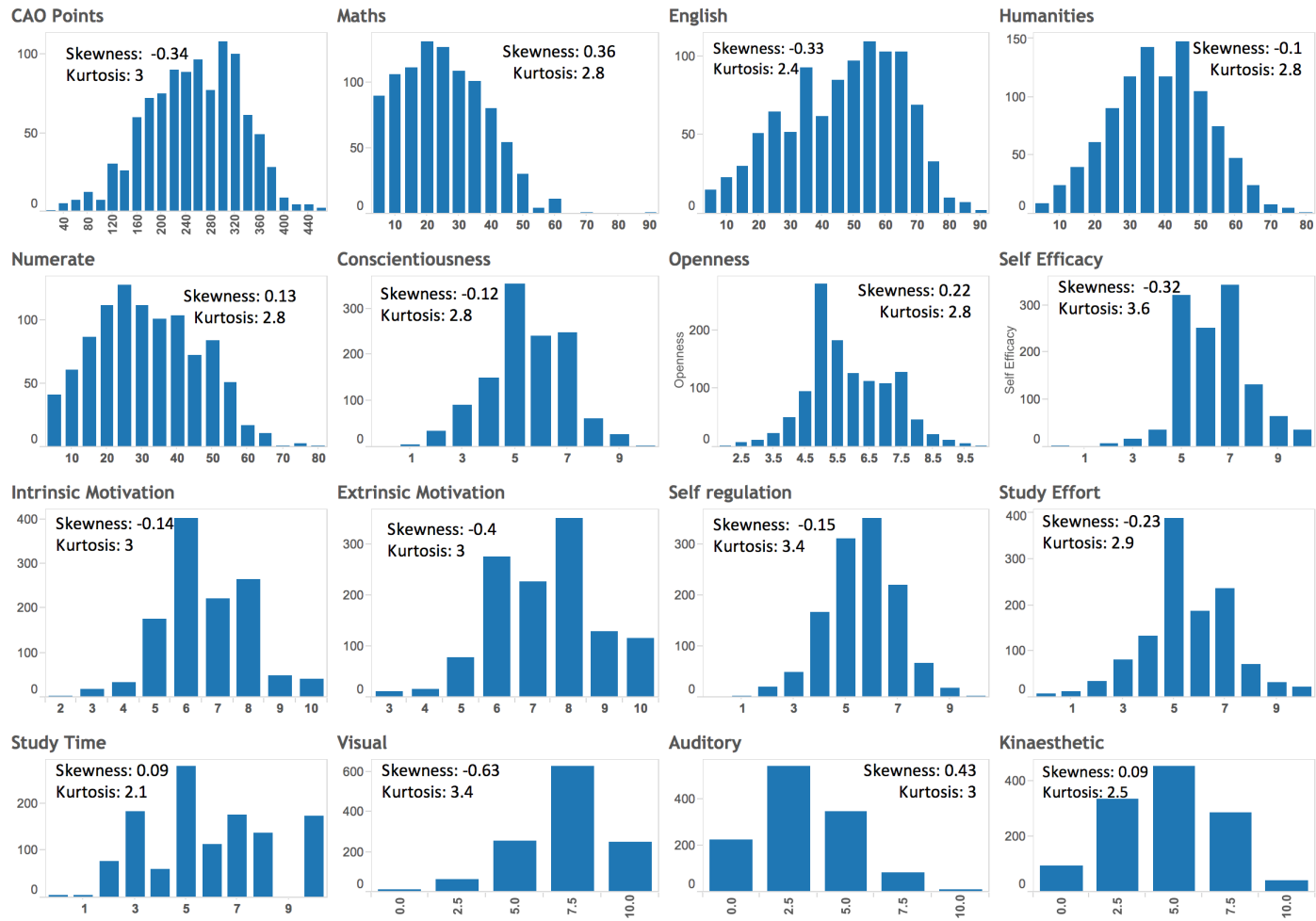


Figure B.1: Histograms for dataset attributes

Appendix C

Extract of code used

Include R code using verbatim

C.1 Bootstrap Confidence Intervals using the Bias corrected and accelerated method

```
%i.e. print as is, do not interpret symbols as latex directives.

library(boot)

#Read dataset
data<-read.csv("FinalDataset.csv",header=TRUE)

#optionally, calculate the bootstrap statistics on a subgroup within the dataset. Below defines subgroups for male and female:
#1 Males:
data<-data[data$gender==0,]
#2: Females
data<-data[data$gender==1,]

#####
#Select prior academic performance factors for students with leaving cert results
CAO<-data[data$CAOPoints>0,c("GPA","CAOPoints","MethodicalAverage","HumanitiesAverage","Maths","English","AppliedAverage","age")]

#Verify the correct number of rows are selected:
nrow(CAO)

#Define a function for each correlation required
fCAO <- function(CAO, i) cor(CAO[i, 1], CAO[i, 2])
fMaths <- function(CAO, i) cor(CAO[i, 1], CAO[i, 5])
fEng <- function(CAO, i) cor(CAO[i, 1], CAO[i, 6])
fNurate <- function(CAO, i) cor(CAO[i, 1], CAO[i, 3])
fHum <- function(CAO, i) cor(CAO[i, 1], CAO[i, 4])
CAOapplied<-CAO[CAO$AppliedAverage>0,]
fApp <- function(CAOapplied, i) cor(CAOapplied[i, 1], CAOapplied[i, 7])

#Generate 1999 bootstrap statistics using correlation functions defined above
resultCAO<-boot(data = CAO, statistic = fCAO, R = 1999)
resultMaths<-boot(data = CAO, statistic = fMaths, R = 1999)
resultEng<-boot(data = CAO, statistic = fEng, R = 1999)
resultNurate<-boot(data = CAO, statistic = fNurate, R = 1999)
resultHum<-boot(data = CAO, statistic = fHum, R = 1999)
resultApp<-boot(data = CAOapplied, statistic = fApp, R = 1999)

#Calculate BCa confidence intervals, default is 95% confidence interval (conf=0.95)
boot.ci(resultCAO, index=1, type="bca")
boot.ci(resultMaths, index=1, type="bca")
boot.ci(resultEng, index=1, type="bca")
```


C. EXTRACT OF CODE USED

```
boot.ci(resultNumerator, index=1, type="bca")
boot.ci(resultHum, index=1, type="bca")
boot.ci(resultApp, index=1, type="bca")

#check other confidence intervals, for example
boot.ci(resultApp, index=1, type="bca", conf=0.999)
boot.ci(resultApp, index=1, type="bca", conf=0.99)
```